```
//#define TFT_MISO 19
//#define TFT_MOSI 23
//#define TFT_SCLK 18
//#define TFT_CS   15  // Chip select control pin
//#define TFT_DC    2  // Data Command control pin
//#define TFT_RST   4  // Reset pin (could connect to RST pin)
//#define TFT_RST  -1  // Set TFT_RST to -1 if display RESET is connected to ESP32 board RST

// For ESP32 Dev board (only tested with GC9A01 display)
// The hardware SPI can be mapped to any pins

#define TFT_MOSI 21 // In some display driver board, it might be written as "SDA" and so on.
#define TFT_SCLK 1
#define TFT_CS   2// Chip select control pin
#define TFT_DC   3 // Data Command control pin
#define TFT_RST  4  // Reset pin (could connect to Arduino RESET pin)
#define TFT_BL   5 // LED back-light

//#define TOUCH_CS 21     // Chip select pin (T_CS) of touch screen

//#define TFT_WR 22    // Write strobe for modified Raspberry Pi TFT only

// For the M5Stack module use these #define lines
//#define TFT_MISO 19
//#define TFT_MOSI 23
//#define TFT_SCLK 18
//#define TFT_CS   14  // Chip select control pin
//#define TFT_DC   27  // Data Command control pin
//#define TFT_RST  33  // Reset pin (could connect to Arduino RESET pin)
//#define TFT_BL   32  // LED back-light (required for M5Stack)

// ######       EDIT THE PINs BELOW TO SUIT YOUR ESP32 PARALLEL TFT SETUP        ######

// The library supports 8 bit parallel TFTs with the ESP32, the pin
// selection below is compatible with ESP32 boards in UNO format.
// Wemos D32 boards need to be modified, see diagram in Tools folder.
// Only ILI9481 and ILI9341 based displays have been tested!
```

Fig. 3: Setting the pins of display driver

file for the library and uncomment the GC9A01_Driver. Set the MISO, MOSI, CS, and DC pins in the configuration file. Refer to Figs. 2 and 3 for settings.

Now, two codes need to be written: one for the touchpad mouse and the other for the keyboard.

Code for touchpad mouse. For the mouse, include the mouse HID library and define the I2C pins for the display. Any pin on the Indusboard can be used as the I2C pin. Here, the pins 5 and 6 are used as SDA and SCL for I2C pins. In the code, map the touch points with mouse movements and use gestures like left swipe, right swipe, single tap, and double tap for mouse clicks. Refer to Figs. 4 and 5 for code snippets.

Code for keyboard. Include the HID library for Indusboard and set the I2C pins of the touch display driver. Any pin on the Indusboard can be configured as an I2C pin.

sketch_feb6b.ino

```
#include "USB.h"
#include "USBHIDMouse.h"
#include "USBHIDKeyboard.h"
USBHIDMouse Mouse;
USBHIDKeyboard Keyboard;


CST816S touch(5, 6, 10, 7); // sda, scl, rst, irq
 int movex=0;
 int movey=0;
void setup() {
  Serial.begin(115200);

  touch.begin();
   Mouse.begin();
  Keyboard.begin();

  Serial.print(touch.data.version);
  Serial.print("\t");
  Serial.print(touch.data.versionInfo[0]);
  Serial.print("-");
  Serial.print(touch.data.versionInfo[1]);
  Serial.print("-");
  Serial.println(touch.data.versionInfo[2]);
 Keyboard.write('u');

}

```

Fig. 4: Code snippet for setting I2C pins

```
  Serial.print("-");
  Serial.println(touch.data.versionInfo[2]);
 Keyboard.write('u');

}


void loop() {

  if (touch.available()) {
    Serial.print(touch.gesture());
    Serial.print("\t");
    Serial.print(touch.data.points);
    Serial.print("\t");
    Serial.print(touch.data.event);
    Serial.print("\t");
    Serial.print(touch.data.x);
    Serial.print("\t");
    Serial.println(touch.data.y);
     movex=(map(touch.data.x,0,244,-40,40));
     movey=(map(touch.data.y,0,244,-40,40));
    Mouse.move(movex, movey);
    if(touch.gesture() == "SINGLE CLICK"){
      Mouse.click(MOUSE_LEFT);

    }
  }



}
```

Fig. 5: Code snippet for mouse HID